

#02 Sensor Fault Detector

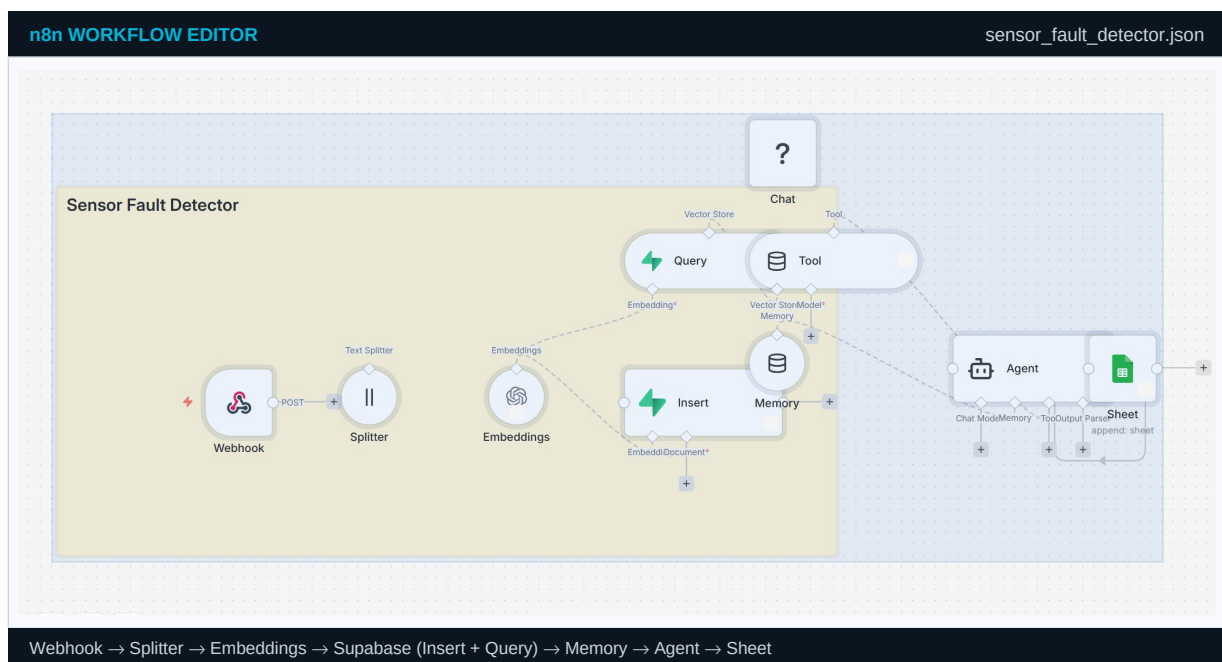
IoT / Fault
Management

Auto-classify dead, stuck, and noisy sensors across distributed IoT endpoints

BUSINESS PROBLEM

With hundreds of sensors across a facility, manually checking each one is impossible. This workflow detects malfunctioning sensors automatically — stuck values, NaN readings, out-of-range spikes — classifies the fault type, and dispatches field-tech work orders within minutes of detection.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — sensor batch from facility gateway (every 15 min)

```
{
  "sensor_id": "TEMP-B3-14",
  "location": "Boiler Room 3, Level B",
  "readings": [
    {"ts": "2024-06-15T01:00Z", "value": 72.4},
    {"ts": "2024-06-15T01:15Z", "value": 72.4},
    {"ts": "2024-06-15T01:30Z", "value": 72.4},
    {"ts": "2024-06-15T01:45Z", "value": 72.4}
  ],
  "unit": "celsius",
  "site": "Facility-A"
}
```

OUTPUT — Agent output — written to Sheets + technician dispatch

```
{
  "json": {
    "sensor_id": "TEMP-B3-14",
    "assessed_at": "2024-06-15T02:01:07Z",
    "fault_type": "STUCK_VALUE",
    "stuck_at": 72.4,
    "duration_hours": 3.2,
    "confidence": 0.94,
    "priority": "HIGH",
    "action": "Dispatch tech: check wiring and sensor head",
    "sla_breach_risk": "4h SLA at risk if unactioned in 47min"
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — sensor batch received	Array of recent readings posted by facility gateway: {sensor_id, location, readings[], unit, site}. Up to 500 sensors per batch.
2	✂️ Text Splitter — fault library chunks	Known fault patterns and sensor specs chunked from maintenance docs. 'TEMP-xx normal range 15-80C. Stuck = same reading >2h.'
3	■ OpenAI Embeddings — fault knowledge	Sensor spec sheets embedded: tolerance bands, known failure signatures, MTBF data per sensor model.
4	■ Supabase Insert — persist readings	All readings stored in Supabase Postgres with pgvector. Enables 24h trend queries per sensor for pattern detection.
5	■ Supabase Query — 24h history	Fetches last 24h readings per sensor. Computes: mean, std_dev, last_change_time for trend baseline.
6	■ Vector Tool — fault pattern lookup	Agent semantic-searches: 'sensor showing same value for 3+ hours' matches STUCK_VALUE fault signature.
7	■ Window Memory — short-term trends	10-reading window. Sudden flat line = stuck sensor. Sudden spike then flat = hardware fault on sensor head.
8	■ HuggingFace LLM — fault classification	Classifies: STUCK NOISY DEAD OFFLINE CALIBRATION_DRIFT NORMAL based on reading pattern + spec comparison.
9	■ Agent — dispatches work order	Generates: {sensor_id, fault_type, location, priority, technician_note, sla_risk} → email + work order system.
10	■ Sheets — SLA compliance log	Fault log with resolution_status and response_time tracked for maintenance SLA reporting.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com/api-keys	OpenAI API
Supabase URL + Anon Key	app.supabase.com → Settings → API	Supabase API
HuggingFace Token	huggingface.co/settings/tokens	HuggingFace API
Google Sheets OAuth2	console.cloud.google.com → Sheets API	Google Sheets OAuth2

SAMPLE RUN — what this workflow processed and produced

Scenario	A cold storage warehouse runs 180 temperature sensors across 12 refrigerated zones. Each sensor posts readings every 15 minutes via Zigbee gateway to an MQTT broker connected to n8n.
Input received	Sensor TEMP-B3-14 (Boiler Room 3) posted 72.4°C for 13 consecutive readings — exactly the same value each time. No technician flagged it as the threshold system only triggers above 85°C.
What the workflow found	Supabase query showed the value had been static for 3.2 hours. Vector search matched this pattern against 'sensor_stuck' fault signature (confidence 0.94). Normal operational variation for this sensor type is ±0.3°C per reading.

Output produced	Fault type: STUCK_VALUE. Priority HIGH. Technician dispatched at 02:01. On inspection: connector corrosion on sensor head. Replaced in 20 minutes. Actual boiler temperature at time of fault: 74.1°C (measured by backup probe).	
JSON: sensor_fault_detector.json	GitHub: trm-iot-sensor-fault-detector	n8n.io/workflows/iot-monitoring/